

Research Note — LLD Practice Platform

1. The learner problem

DSA practice has a tight loop: read, code, run tests, instant verdict. LLD has no such loop.

- **No single correct answer.** Designs trade extensibility against simplicity; there is nothing to diff against.
- **Blank-page paralysis.** Learners draw large class diagrams before isolating responsibilities and invariants.
- **Ungrounded feedback.** AI feedback on LLD now exists, but it is per-session, not tied to the learner's own text, not pinned to a versioned rubric, and not aggregated across problems.
- **No longitudinal signal.** Learners cannot tell whether habits (God classes, anemic models, missing abstractions) are improving.

2. Competitive landscape

Rows 1–4 are based on public marketing and free-tier pages; premium feedback internals are paywalled and unverified.

#	Tool	Workflow	Submission	Feedback	Loop	Gap
1	Hello Interview	9 LLD problems, ~35 min, requirements→design→extensibility	Free-form per step	Instant AI, described as rubric-based	Per-problem history	Rubric not exposed or versioned; no cross-problem aggregation; not tied to quoted spans
2	algomaster.io	45 problems, 9 sections	Undocumented	"AI powered evaluation"	Section progression	Single opaque pass; cannot tell which criticisms are mechanically certain
3	Low Level Design Mastery	43 problems; pick→diagram→code→AI review	Diagram + code, 6 languages	AI review at end	XP, levels, drills	Progression is volume-based, not diagnosis-based; repeat mistakes still level up
4	lldproblems.com	100+ machine-coding problems; gated unlocks	Chat simulation	Conversational interviewer	Mastery tiers	Output not structured into stable dimensions, so nothing is trackable
5	Educative / Grokking LLD	Chaptered lessons on OOD and UML	None (read-only)	Worked solutions to self-compare	Completion %	No evaluation of the learner's design; self-scoring is systematically generous
6	AlgoExpert / SystemsExpert	25 modules, 13 design questions, 38 videos	Notes in a workspace	Video solutions	Quiz scores	Workspace records the design but nothing reads it
7	Exercism	Solve, auto-tests, then human mentor	Code	Tests, then mentor hints; learners resubmit iterations	Iteration history, mentor continuity	Closest real loop, but hours-to-days latency and no LLD coverage at all
8	ChatGPT	Paste design, ask for critique	Prose	Fluent, immediate, ungrounded	None	No fixed rubric, no citations, invents strengths, no memory of prior attempts

3. Market gaps

1. **Not grounded in the learner's words.** Rows 1–4 critique without citing the span they react to; a learner who disagrees cannot check. → `sourcePath` citations plus explicit absence references.
2. **Scores not comparable over time.** Rows 1, 3, 4 track activity, not per-dimension movement against a pinned rubric. Without `rubricVersion`, two 3.5s are different measurements. → versioned rubric plus per-dimension deltas.
3. **Facts and opinions indistinguishable.** Rows 2, 3, 4, 8 return one verdict. "11 methods" is checkable; "feels forced" is judgement. → deterministic/LLM split with contributor badges.
4. **Nothing reports the repeated mistake.** Row 7 has iteration history but no design coverage; rows 5, 6, 8 have no loop. → recurring-weakness aggregation across problems.

5. **The one real human loop is too slow.** Row 7 shows learners iterate when feedback is specific, but hours of latency breaks practice rhythm. → sub-15s hybrid evaluation, human review as a documented amendment path.

4. Product direction

- **Canonical structured spec** (`assumptions` , `entities` , `relationships` , `interfaces` , `tradeoffs` , `extensibility`) gives deterministic checks real anchors and grounds LLM judgement.
- **Hybrid evaluation:** zero-cost deterministic heuristics plus a temperature-0, Zod-validated LLM evaluator, merged by confidence weighting with full finding provenance.
- **Honest citations:** every finding carries a path into the learner's submission, or an explicit absence notice. No fabricated quotes.
- **Longitudinal loop:** per-dimension deltas between attempts plus cross-problem recurring-weakness aggregation.